MANET C4ISTAR Planning

Quick briefing

A mobile ad hoc network (MANET) lets mobile nodes form and adapt wireless paths without a permanent infrastructure. Plan the services those paths must carry and verify that each service works under exercise conditions.

Know the facts

Mobile nodes · Nodes can forward traffic for one another across multiple hops where a usable path exists.

Self-healing · The network can adapt to a topology change by using an available alternate path. A disconnected group may still lose service.

Traffic and capacity · Unicast serves an individual destination, while broadcast reaches a broader audience. Quality of service helps manage competing traffic within available capacity.

Diagnostics · Received signal strength and signal-to-noise ratio measure different conditions. Review them alongside service and network evidence.

Fault handling · Identify and isolate the fault, carry out the approved repair and verify that the affected service has recovered.

Integration and protection · An integrator can connect legacy and newer services. Verify each needed service separately and preserve the controls on what may cross the boundary.

Understand the reasoning

A network can reroute and still struggle with heavy updates. Judge restoration using the service the user needs, including its delay or loss.

A connected node and a successful voice test give limited evidence about data exchange. Each required service has its own traffic and compatibility demands.

Philippine Army classroom exercise

During a Philippine Army classroom simulation, one moving node loses its usual path. The network reroutes traffic and voice works, but map updates arrive too late for the exercise's movement decision.

Decision · What should the team verify before it declares the network restored?

Worked analysis · Confirm that the receiving team gets a usable map update within the exercise's decision window. Compare the data path and traffic behavior before and after the reroute, and check whether the delay affects only that service. If capacity is limited, prioritize essential exchanges and repeat the same test.

How to practice

1 · Define a success check for each required classroom service, such as intelligible voice or a timely map update.

2 · Record which simulated nodes and services work before the instructor introduces a path or interface fault.

3 · Use the authorized diagnostic view to compare the affected path and services, then isolate the likely fault.

4 · Repeat the original service checks after the instructor-approved correction and record the outcome.

Check yourself

What must exist for self-healing to restore connectivity after a path breaks? · A usable alternate path between the affected nodes.

Why should repair verification go beyond seeing that a node is connected? · Connectivity alone leaves the actual delivery and performance of the required service unverified.

Course source

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